

Detecting Cloned Voices on a Phone Call

We are not proposing to build a detector that works. We are proposing to measure exactly where detection breaks, on the one axis this field has never measured: latency.

TEAM

[names]

DEPLOYMENT TARGET

On-device app, live call audio

PRIMARY CORPUS

ASVspoof 2019 LA → 2021 LA

DATE

September 2026

Ninety seconds of public audio is enough.

- An attacker scrapes a voice from a podcast, a conference talk, a voicemail.
- A zero-shot cloning model reproduces it. **No training, no GPU farm, minutes of work.**
- The call arrives at a family member, an employee, or a bank's phone line.
- The victim hears a voice they recognise and acts on it.

THE QUESTION THIS TALK ANSWERS

A human hears a familiar voice and believes it. **Can a machine, listening to the same call, tell that it is synthetic?**

The literature says this is **solved**.

BEST REPORTED

1.23%

RawGAT-ST equal error rate on ASVspoof
2019 LA evaluation.

BENCHMARK MATURITY

4

Biennial ASVspoof challenges since 2015. 54
teams submitted to the 2021 evaluation
alone.

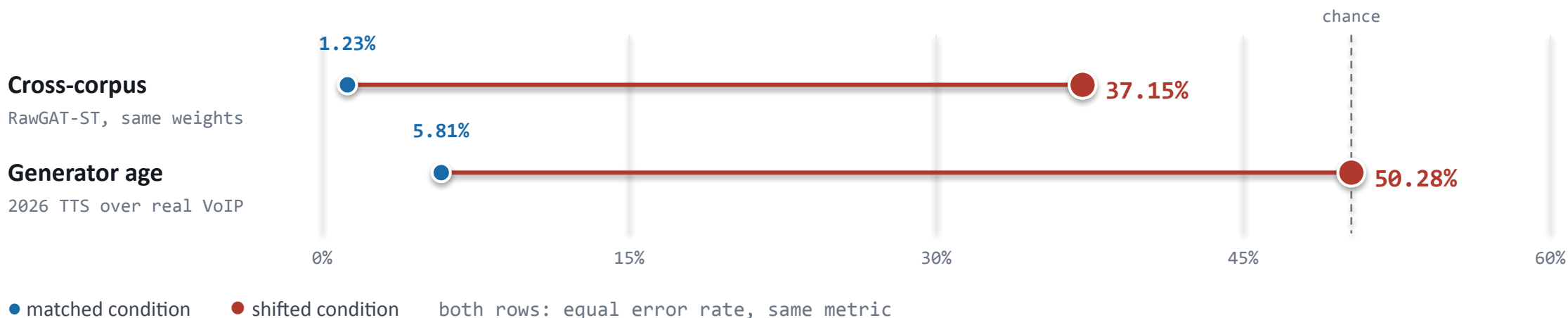
ARCHITECTURES COMPARED

12

Re-implemented under one protocol by
Müller et al. Raw-waveform models win
consistently.

On paper, audio deepfake detection is a benchmark with a clear winner and a long tail of incremental improvement. **Three independent results say otherwise.**

Move one variable and it goes to **chance**.



THIRD COLLAPSE — DIFFERENT METRIC, SO STATED SEPARATELY

Seven cheap DSP transforms raise the share of spoofs a countermeasure accepts from roughly 2% to **12–66%**. It still works over a real Twilio call into Amazon Connect at 8 kHz. Kassis & Hengartner, IEEE S&P 2023

Three different groups, three different failure modes, none of them marginal. Training on more ASVspoof data does not help: 33.9% → 33.1%, because every split derives from the same VCTK recordings. Generator-age row is a recent preprint (RTCFake) and is spoken as such.

The detector is not listening to the **voice**.

SILENCE ALONE

85%

Accuracy from a classifier fed **only the duration of the leading silence**. Nothing else.

Müller et al.

ATTACK ON SILENCE ONLY

82.2%

Attack success from perturbing **only background noise and mute segments**, never the voice.

SiFDetectCracker, ACM MM 2023

WHERE IT ATTENDS

0.5–0.6 kHz

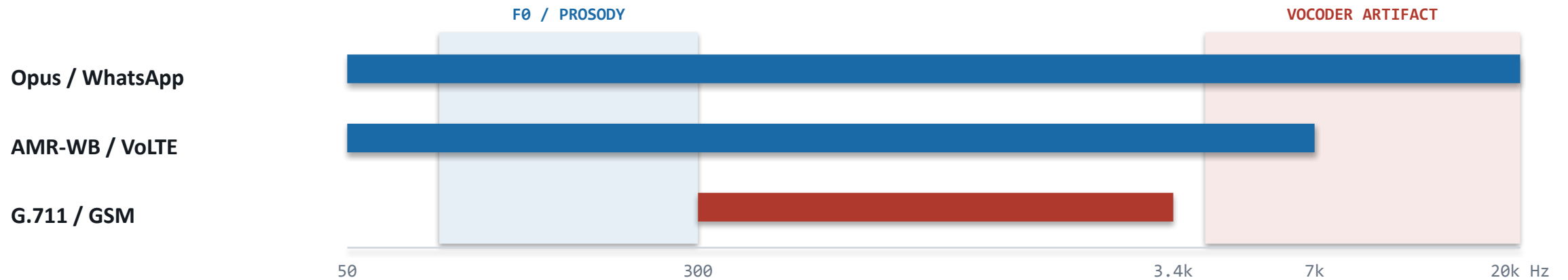
SHAP analysis: classifiers attend to **non-speech intervals** and one narrow low band.

arXiv:2110.03309

WHY THIS MATTERS FOR US SPECIFICALLY

A meaningful share of published performance is a **dataset artifact**. An attacker deletes the silence and it is gone. And a live phone call has no clean studio silence to begin with, so that cue transfers nothing to our deployment.

What actually survives a phone line?



THE TWIST, AND IT CHANGES OUR THREAT MODEL

Narrowband does not leave nothing. Low-pass filtering to the telephone band actually **improves** codec robustness, ~25% relative, and the **F0 sub-band alone reaches 1.15% EER**. What survives a phone line is **prosody** — exactly what the cheap attacks on the previous slide manipulate.

arXiv:2211.06546, arXiv:2208.01214

Nobody has measured the thing the brief asks for.

- Every ASVspoof score is **utterance-level and offline**, computed over a complete recording.
- **Zero** papers in this literature report milliseconds per decision.
- Every top-5 system in ASVspoof 2021 was an **ensemble fused by score averaging** — structurally incompatible with a live-call budget.
- There is **no agreed metric** for a streaming decision. It has to be defined before any number means anything.

PUBLISHED GUIDANCE

“Use the full recording, do not cut it into short chunks.” Average EER on ASVspoof: **9.85%** full vs **18.89%** at 4 s.

But on realistic audio the advice reverses. RawNet2 In-the-Wild: **37.82%** full vs **33.94%** at 4 s. Short windows **win**.

The recommendation against short windows appears to be an artifact of the clean benchmark. **Real-time may cost far less than the field assumes, and that is directly testable.**

An app on the callee's phone, listening live.

WHAT THIS BUYS US

A generous latency budget. The alarm is read by a human who needs seconds to react, not an authentication gate that must answer in 200 ms.

A clear cost asymmetry. An app that cries wolf gets uninstalled, so **false alarms matter more than misses**. That sets our operating point, not EER.

WHAT WE MUST NOT HIDE

Android will not give a third-party app the call audio. The VOICE_CALL source has needed privileged access since Android 10. iOS is closed outright.

The honest capture path is **speakerphone into the microphone**, which adds an acoustic channel on top of the codec — measurable damage: W2V2-AASIST **4.7% → 18.2%** EER under replay.

WE TREAT THIS AS A FINDING, NOT AN OBSTACLE

The literature evaluates on clean file input. **No published number accounts for the acoustic re-capture an on-device app actually forces.** We will measure that stage explicitly and report it as its own condition.

One window is not a decision.

THE TRAP WE HAVE TO AVOID

Alarm whenever any window crosses a threshold, at 5% per-window false-alarm rate and 2-second windows, and a **60-second call false-alarms 79% of the time**. Per-window error rate is the wrong headline number.

PRIMARY RULE

Sequential evidence accumulation

Accumulate a calibrated log-likelihood ratio across windows and alarm when it crosses a boundary — Wald's sequential probability ratio test, with a raise/clear pair for hysteresis and a 3-second evidence floor.

Why this one: it is the classical optimum for “decide as soon as the evidence is sufficient”, so the time-to-decision versus error-rate curve is its natural output. Our latency metric is inherited from sequential detection theory rather than invented.

BASELINE TO BEAT

k-of-n consecutive windows

Simple, explainable, and a fair comparator. We report both.

Reported quantities — all three, always: false alarms **per call-minute**, detection rate, and time-to-detection at median and 90th percentile.

Honest caveat we will state in the report: adjacent windows are correlated, which violates the independence assumption. We calibrate an effective sample size empirically rather than pretending the assumption holds.

What we will build.



Every codec above is already available in the ffmpeg build on our machine, so the harness carries no setup cost. Detector training fits on one 16 GB laptop GPU.

Two numbers nobody has produced.

CONTRIBUTION 1

A latency / accuracy curve for streaming detection over telephony

Unmeasured in this field. Producing it requires first **defining the streaming decision metric**, which does not currently exist, and that definition is itself part of the contribution.

CONTRIBUTION 2

The false-alarm rate on genuine human speech carried over GSM

GSM Full Rate is **itself an LPC vocoder**. Real speech arrives already vocoded, so the detector's core cue becomes ambiguous and may fire on honest callers.

The ASVspoof organisers raised this in 2017. It has never been quantified.

We claim the **measurement**, not the idea. Contribution 2 was predicted by the challenge organisers and we cite them for it.

Three rules so we do not fool ourselves.

- 1 Every error rate is reported twice**, with and without VAD silence trimming. Given what the silence cue is worth, a number without this pair is not evidence about voice authenticity, and any expert in the room can say so.
- 2 Per generator and per condition, never pooled.** Pooling hides the cross-corpus term, which is the largest term we found.
- 3 No self-generated sample ever enters training.** This removes the circularity objection outright. The precedent is ASVspoof's own design: attacks A07–A19 appear in evaluation only, and the organisers sourced them externally so that countermeasure developers had no prior knowledge.

KNOWN-ATTACK CONDITIONS LOOK WONDERFUL AND MEAN NOTHING

With attacks known in advance, 12 of 16 ASVspoof 2015 systems scored under 1% EER, the best at **0.003%**. A single unseen attack took the same systems to **8–46%**.

Twelve weeks, and where it will hurt.

PHASE	WEEKS	MILESTONE	RISK
Baseline	2–4	RawNet2 reproduces published clean-condition EER	Low. Proves the setup, nothing more
Degradation harness	5–7	Per-codec error rates across the bandwidth ladder	Historically overruns
Streaming	8–9	Latency curve and the SPRT decision rule	Engineering-heavy, and the core contribution
Probe & integration	10–11	Modern-attack probe, GSM false-alarm measurement	May yield a flat result with no gradient
Write-up	12	Report and defence	Low

Compute is a single laptop GPU, 16 GB, which is sufficient but **serialises our experiments** — roughly 1.5 to 3 hours per training run. Phases 5–7 and 8–9 are where we expect to slip, and we are saying so now rather than in week 11.

What we will claim, and what we will not.

WE WILL NOT CLAIM

That we have built a reliable authentication gate for voice.

That a single error rate from one corpus describes field performance.

That the problem is solved, or that we solved it.

WE WILL CLAIM

A measured map of **where detection breaks**: by codec, by bandwidth, by corpus, by window length.

The first **latency / accuracy curve** for streaming detection over a telephony channel.

The first **false-alarm figure for genuine speech over GSM**.

THE POSITION WE WILL DEFEND

Audio presentation-attack detection is a **risk signal with a measured error rate**, fused with caller-ID and signalling checks, account context and challenge-response. **It raises attacker cost. It does not authenticate anyone by itself.**

Questions we expect.

QUESTION

ANSWER

Why not a wav2vec2 / SSL front-end?

It is the most robust under attack — 1.89% → 11.6% where SSNET goes to 61.81%. It is also the heaviest, which is precisely the latency trade-off we propose to measure.

Why not just train on more data?

Measured, and it does not work: 33.9% → 33.1%. Every ASVspoof split derives from the same VCTK recordings.

Why not ASVspoof 5?

Newer, and it does add adversarial attacks. Still 43.71% EER against modern generators over real VoIP. Same failure.

Is this not just reproducing known results?

Latency is unmeasured. The GSM false-alarm rate has been unquantified since 2017. AMR codecs appear in no ASVspoof edition.

Is your codec simulation realistic?

We validate against ASVspoof 2021 LA, which is real VoIP and PSTN transmission. The organisers themselves warn that simulated data misled the replay task.

Why not build the attack side properly?

Cost, ethics, and redundancy. Public corpora already hold more generator diversity than we could produce, and cloning third-party voices is not defensible here.